smaller than the pupil gives less blur than the pupil, so the view is clearer.

For the patch diameter, Fig. 20-5-1 gives the lens formula:

$$\frac{1}{a}+\frac{1}{b}=\frac{1}{f} \tag{1}$$

The focal length is:

$$f=\frac{ab}{a+b} \tag{2}$$

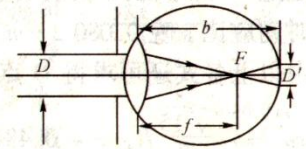


Fig. 20-5-2

In Fig. 20-5-2, parallel rays from a distant axial point pass through a hole of diameter D, focus at F and form a patch D' wide on the retina, as Fig. 20-5-2 shows.

Geometry and (2) give patch diameter D':

$$D'=\frac{b-f}{f}D=\frac{b}{a}D \tag{3}$$

By (3), smaller D gives smaller D' in geometric optics.

2. Parallel rays through the hole and lens form a round retinal diffraction spot of diameter D''. The given approximation yields:

$$D''=2b\Delta\theta=2.44\frac{\lambda b}{D} \tag{4}$$

By (4), smaller D gives larger D'': a blurrier point. So do not make the hole too small. The optimum diameter D makes the geometric and diffraction spots equally large:

$$D'=D'' \tag{5}$$

The image is clearest. For hole diameter D_0:

$$\frac{b}{a}D_0=2.44\frac{\lambda b}{D_0} \tag{6}$$

$$D_0=\sqrt{2.44\lambda a} \tag{7}$$

Substitute $\lambda=600\times10^{-9}$ m and $a=0.20$ m:

$$D_0=0.54\times10^{-3}\ \text{m} \tag{8}$$

6.

1. A charged cosmic-ray particle has speed v. Geomagnetic Lorentz force gives circular motion:

$$qvB=m\frac{v^2}{R} \tag{1}$$

Here m is relativistic mass, related to rest mass by:

$$m=\frac{m_0}{\sqrt{1-v^2/c^2}} \tag{2}$$

Its kinetic energy is:

$$E_k=mc^2-m_0c^2 \tag{3}$$

From (1):

$$v=\frac{qBRc}{\sqrt{q^2B^2R^2+m_0^2c^2}} \tag{4}$$

Using the data:

$$v=2.90\times10^8\ \text{m}\cdot\text{s}^{-1} \tag{5}$$

From (2), (3):

$$E_k=m_0c^2\left(\frac{1}{\sqrt{1-v^2/c^2}}-1\right) \tag{6}$$

Using (5) and the data:

$$E_k=1.74\times10^{-9}\ \text{J} \tag{7}$$

2. Let p, E be the particle's momentum and energy:

$$p=\frac{m_0v}{\sqrt{1-v^2/c^2}} \tag{8}$$

$$E=\frac{m_0c^2}{\sqrt{1-v^2/c^2}} \tag{9}$$